

12 x 20 Deck with Landing and Stair

Prepared for Alan · 11215 W Center Ave, Lakewood, CO 80226 · September 16, 2026

CHECK OR ACH	FINANCED	MONTHLY
\$59,790	\$63,975 no money down	\$742 6.99% · 10 years



A 12 x 20 deck, 8 feet above grade, wrapping the inside corner of the house: ledgered to the Hardie wall, the north bay on its own beam past the house corner, a 4 x 4 landing at the brick corner and a 13-riser stair down the front to a concrete pad. TimberTech Advanced PVC Vintage in Weathered Teak on 2x10 joists at 12" on center, Cortex screws with Weathered Teak plugs for a hidden face. Impression Rail Express horizontal cable rail in black, 36", with a pipe handrail on the stair. 6x6 posts on nine Diamond Pier footings. The existing deck, pergola and stair come out first.

AT A GLANCE



From the yard



Isometric from the open corner

Frame	3 zones, 20'-0" along the house, 16'-0" at the deepest
Deck	60.0 LF of outer edge · 264 SF · 8'-0" above grade
Zone	N North bay — open end, no house behind it: 8'-0" x 12'-0"
Zone	S South bay — ledgered to the Hardie wall: 8'-0" x 12'-0"
Zone	L Stair landing — 4' bump-out at the brick corner: 4'-0" x 16'-0"
Joists	17 2x10 #1 True Frame Joist SYP GC @ 12" OC
Rims	2-ply 2x10 front and sides and rear (freestanding)
Beam	DROP (2)2x10 SYP N: 7'-11", 2 posts at 1'-0 1/2" / 6'-11 1/2" (worst 1,089 lb)
Beam	DROP (2)2x10 SYP N+S: 15'-11", 3 posts at 1'-0 1/2" / 8'-0" / 14'-11 1/2" (worst 2,261 lb)
Beam	DROP (2)2x10 SYP L: 3'-11", 2 posts at 17'-0 1/2" / 18'-11 1/2" (worst 881 lb)
Posts	9 x 6x6 on DP-50/50 — 9 ordered
Load	50 psf (40 psf live governs)
Decking	TimberTech Vintage Weathered Teak — 84 rows parallel to the house, 3/16" gaps, Cortex
Rail	IRX 36" Black: 7 bays, 9 posts, 44.0 LF
Stair	step:S-L: 13 risers @ 7.38", 12 treads, 5 stringers, 4'-0" wide

WHAT IT IS included in the price

ELEMENT SELECTION

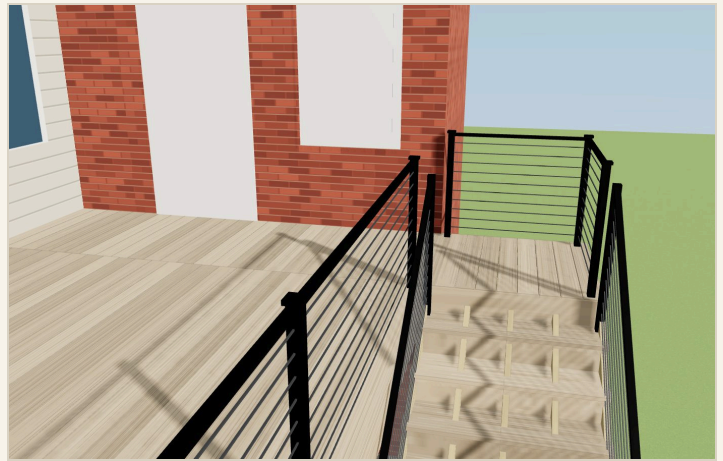
Decking	TimberTech PVC Vintage, Weathered Teak. Square-shoulder boards, 3/16" gaps, matching fascia. Ignition Resistant, Class A, WUI; Limited Lifetime product / 50-yr fade & stain.
Railing	TimberTech IRX cable rail, Black, 36", no bottom rail. 44.0 LF.

ELEMENT SELECTION	
Frame	17 2x10 #1 True Frame Joist SYP GC @ 12" OC; 2-ply 2x10 front and sides and rear (freestanding); DROP (2)2x10 SYP N: 7'-11", 2 posts at 1'-0 1/2" / 6'-11 1/2" (worst 1,089 lb); DROP (2)2x10 SYP N+S: 15'-11", 3 posts at 1'-0 1/2" / 8'-0" / 14'-11 1/2" (worst 2,261 lb); DROP (2)2x10 SYP L: 3'-11", 2 posts at 17'-0 1/2" / 18'-11 1/2" (worst 881 lb).
Foundation	DP-50/50.
Design load	30 psf snow.
Hardware	Simpson ZMAX galvanized connectors. Cortex fasteners with color-matched plugs.

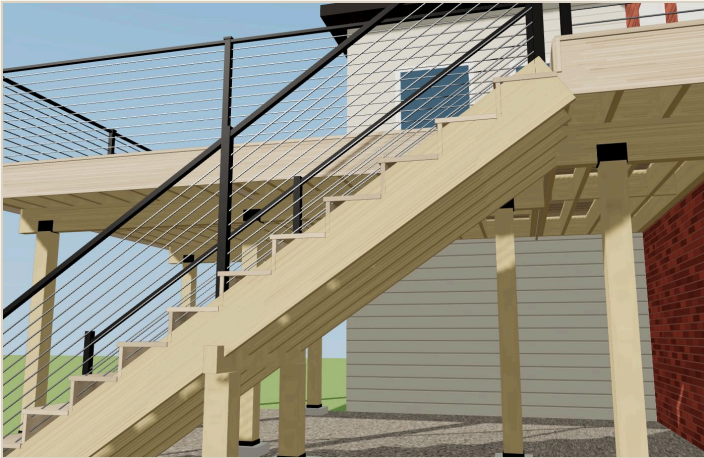
Engineering and permit. GSX provides the engineer and pulls the permit. Engineering and the permit are billed at cost, separate from this price.



From the front corner



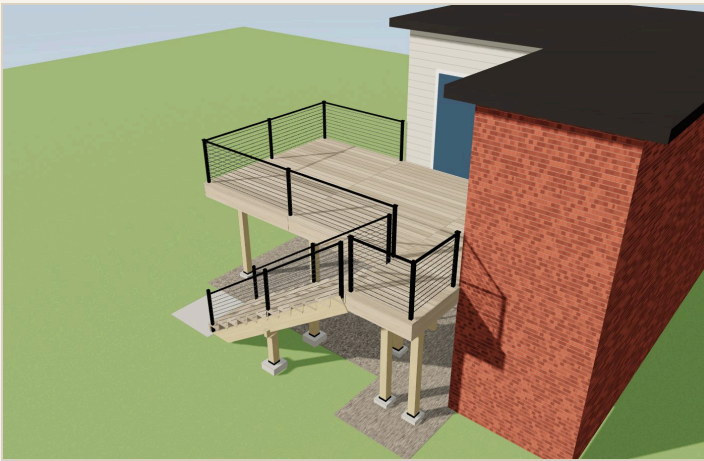
Standing on the deck



Under the deck — posts, beams, footings



Low isometric



Isometric from the stair corner



Isometric from the open corner

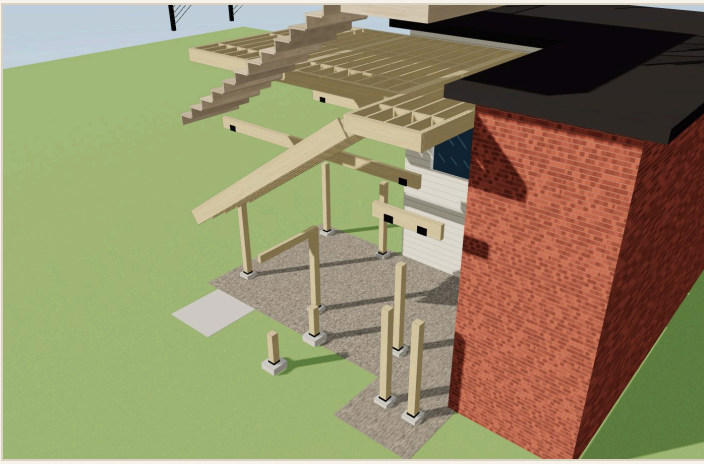


Low isometric

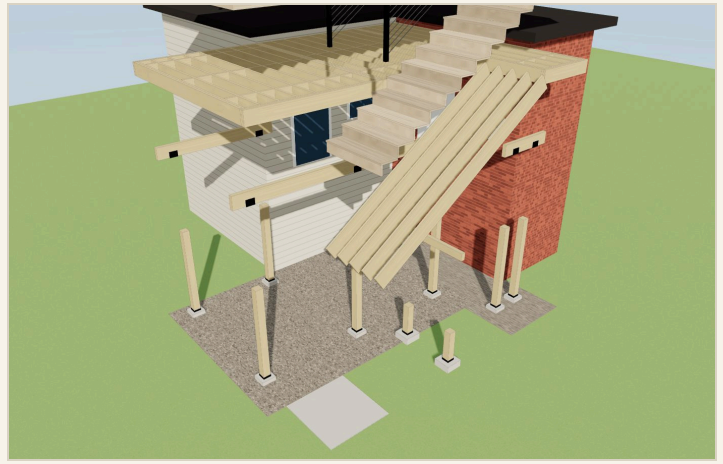


Isometric from under the frame

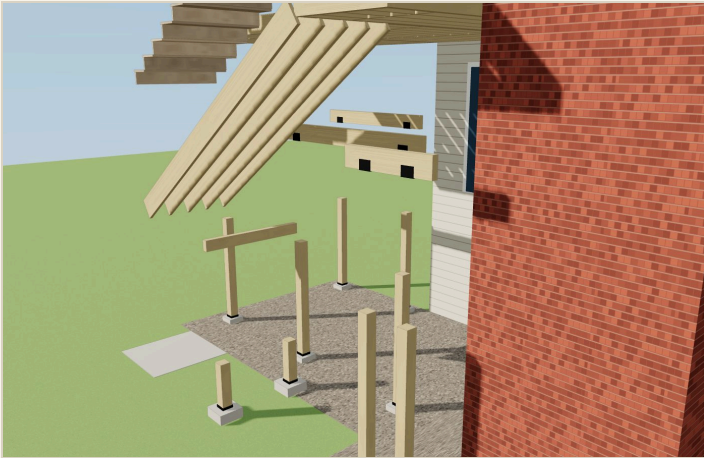
EXPLODED VIEWS every layer pulled apart



Exploded — every layer



Exploded — from the open corner



Exploded — from the front corner



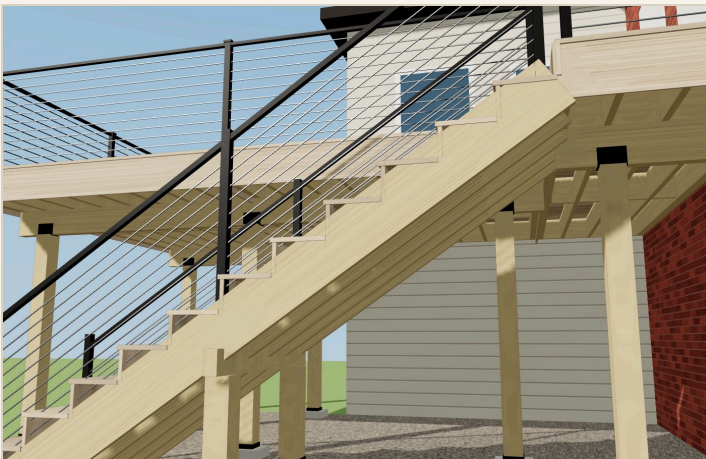
Exploded — low angle



Posts and beams on the footings



Frame only — joists, rims, beams, ledger



Under the deck — posts, beams, footings



Through the rail

G-001 General notes · design criteria

G-001

GENERAL NOTES · DESIGN CRITERIA

Loads, codes, structure summary, sheet index, site — the permit-set cover

Center Ave — 12 x 20 deck with landing and stair · 11215 W Center...

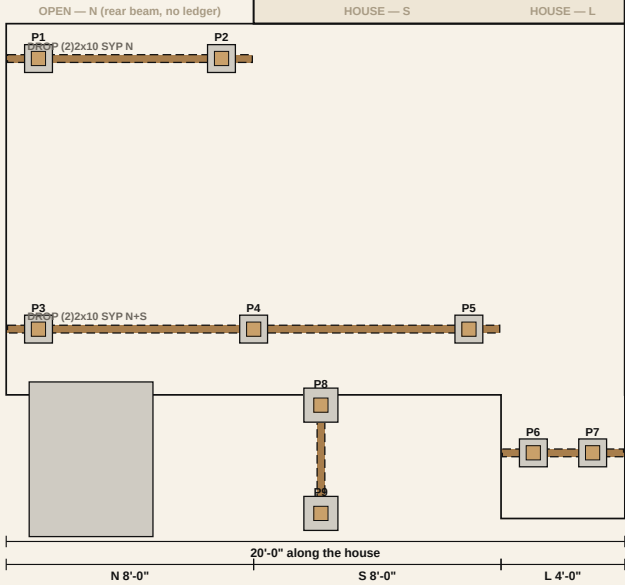
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PROJECT	
Center Ave — 12 x 20 deck with landing and stair	
11215 W Center Ave, Lakewood CO 80226	
Prepared for Alan	
264 SF · 8'-0" above grade · 3 zones	
DESIGN CRITERIA	
Deck live load	40 psf uniform (IRC Table R301.5); guards 200 lb concentrated / 50 plf
Ground snow	30 psf (ground; deck design uses the larger of snow and live) — 40 psf live governs
Dead load	10 psf deck
Roof snow (cover)	—
Design total	50 psf on the deck
Wind	115 mph Vult, Exposure C, Risk Cat. II (ASCE 7-16) — confirm special wind region
Frost depth	3'-0" (footings bear below)
Soil bearing	1,500 psf presumptive (IRC R401.4.1) — no soils report
Seismic	B
Wildfire (WUI)	not in a WUI zone
Codes	2021 IRC as adopted by Lakewood; IRC R507 decks; ASCE 7-16 loads; Tables R507.5 / R507.6 / R507.4 / R507.9
Engineering	Stamped structural set by a Colorado PE, provided by GSX and billed at cost with the permit
STRUCTURE	
Footings	DP-50/50
Posts	9 x 6x6
Beams	DROP (2)2x10 SYP N: 7'-11", 2 posts at 1'-0 1/2" / 6'-11 1/2" (worst 1,089 lb); DROP (2)2x10 SYP N+S: 15'-11", 3 posts at 1'-0 1/2" / 8'-0" / 14'-11 1/2" (worst 2,261 lb); DROP (2)2x10 SYP L: 17 2x10 #1 True Frame Joist SYP GC @ 12" OC
Joists	2-ply 2x10 front and sides and rear (freestanding)
Rims	2x10 ledger x 12'-0" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 13 fasteners
Ledger	
Decking	TimberTech Vintage Weathered Teak — 84 rows parallel to the house, 3/16" gaps, Cortex
Rail	IRX 36" Black: 7 bays, 9 posts, 44.0 LF

SHEET INDEX	
G-001	General notes · design criteria
A-001	Isometric · renders
S-100	Foundation plan
S-101	Framing plan
A-101	Decking plan
A-201	Railing plan + elevation
A-301	Section + front elevation
D-501	Details
T-400	Build steps (build set)
T-700	Order + schedules (takeoff)
SITE	
Lakewood, CO. Existing deck, pergola and stair come out.	

No scale

Design intent per 2021 IRC R507 prescriptive. Field verify. Do not scale.

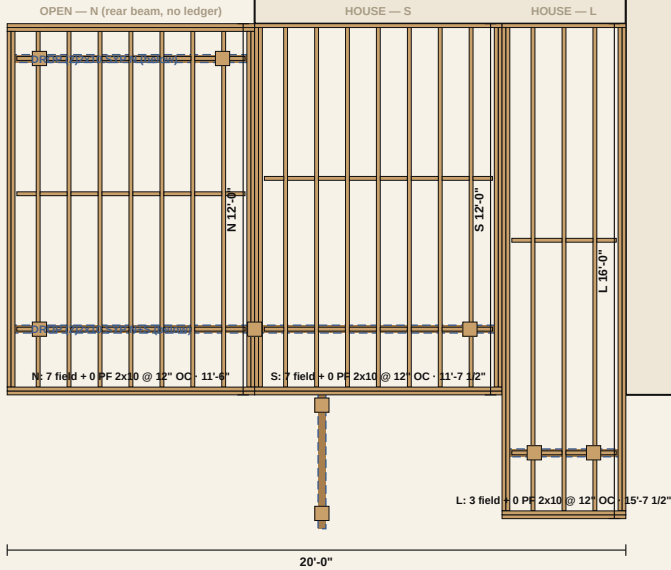


FOOTING SCHEDULE

Diamond Pier DP-50/50 (head + 4 x 50" pins),
3,300 lb allowable

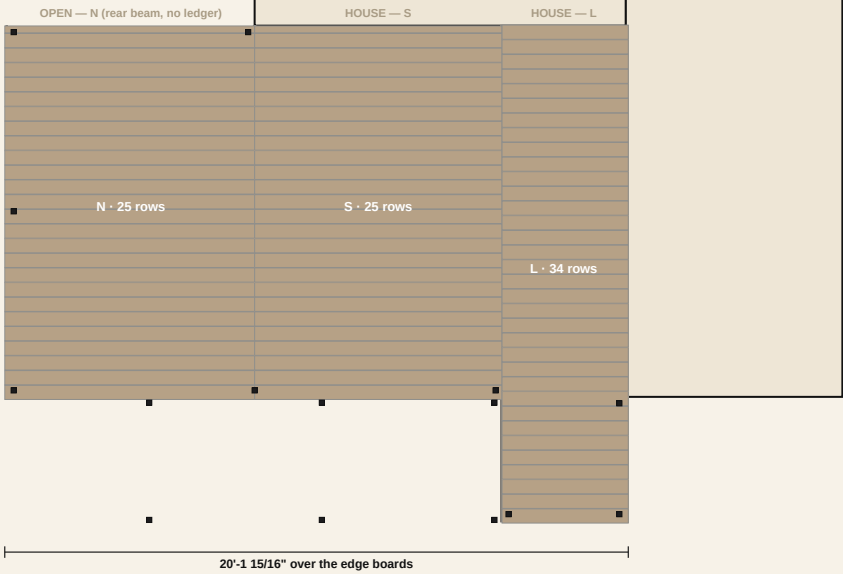
ID	X	Y	ZONE	LOAD
P1	1'-0 1/2"	1'-1 1/2"	N	1,089 lb
P2	6'-11 1/2"	1'-1 1/2"	N	1,089 lb
P3	1'-0 1/2"	9'-10 1/2"	N	1,456 lb
P4	8'-0"	9'-10 1/2"	N	2,261 lb
P5	14'-11 1/2"	9'-10 1/2"	S	1,568 lb
P6	17'-0 1/2"	13'-10 1/2"	L	881 lb
P7	18'-11 1/2"	13'-10 1/2"	L	881 lb
P8	10'-2 1/8"	12'-4"	stair	carrier
P9	10'-2 1/8"	15'-10"	stair	carrier

x from the left frame face facing the house; y from the reference wall.
Frost 3'-0" · soil 1,500 psf · worst post 2,261 lb



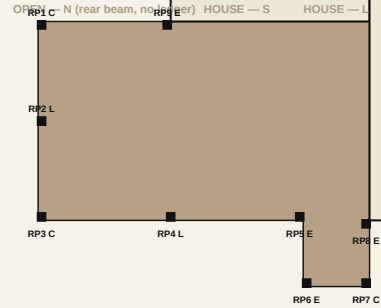
FRAMING NOTES

- 17 2x10 #1 True Frame Joist SYP GC @ 12" OC
- 2-ply 2x10 front and sides and rear (freestanding)
- 2x10 ledger x 12'-0" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 13
- blocking one row over the beam, staggered
- G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)
- DROP (2)2x10 SYP N: 7'-11", 2 posts at 1'-0 1/2" / 6'-11 1/2" (worst 1,089 lb);
- DROP (2)2x10 SYP N+S: 15'-11", 3 posts
- Connectors:
- Simpson 2x10 Shear Face Mount Hanger Zmax: 40
- Simpson 2x10 Double Concealed Face Mount Hanger Zmax: 6
- Simpson Hurricane Ties Z-Max H2.5AZ: 58
- Simpson Adjustable Post Cap 6x6 · Zmax: 3
- Simpson 6x Post Cap Adjustable Zmax: 4
- Simpson 6x Heavy Adjustable Post Cap · Zmax: 4
- Simpson Deck Tension Tie Zmax with 1.5" screws: 4
- Camo 3" PRO-FIT Joist Hanger nail HDG (5 lb box): 1



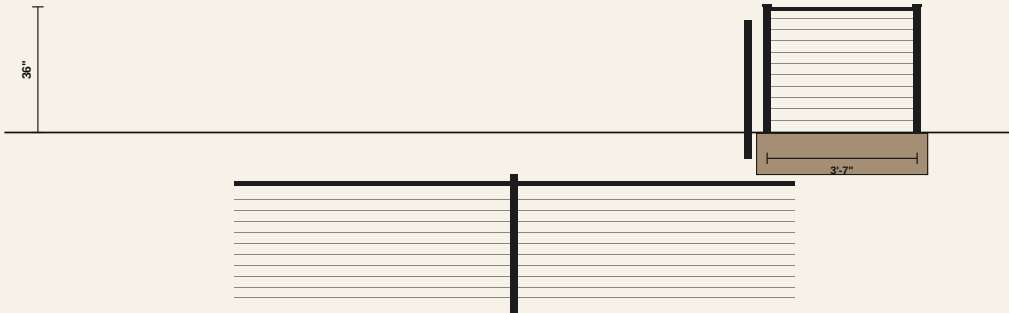
DECKING

Field: TimberTech Vintage Weathered Teak, 5.5" x 1", 3/16" gaps
No picture frame
Boards run parallel to the house; edge overhang 1" over the fascia
Cortex face screws with color-matched plugs: 2 per board at every joist and at both ends of every run, color-matched plugs (1442 field + 0 border/divider)
Rows per zone: N 25 · S 25 · L 34 (rip at the house)



RAILING

IRX 36" Black: 7 bays, 9 posts, 44.0 LF
IRX 36" — 9 posts (3 CORNER, 4 END, 2 LINE), 7 bays at 3'-7" / 5'-9 1/2" / 7'-7" / 7'-9 1/2" CTC (8' kils cut to bay), posts on the divider lines, no bottom rail; HeadLOK 6" x 4 per post
Bays: end:left 5'-9 1/2" · end:left 5'-9 1/2" · front:N+S 7'-9 1/2" · front:N+S 7'-9 1/2" · front:L 3'-7" · end:right 3'-7" · wall:N 7'-7"
Guard 36" min above the deck; openings under 4" (IRC R312). Rail LF for labor = deck-edge LF of actual rail.



A-301 Section + front elevation

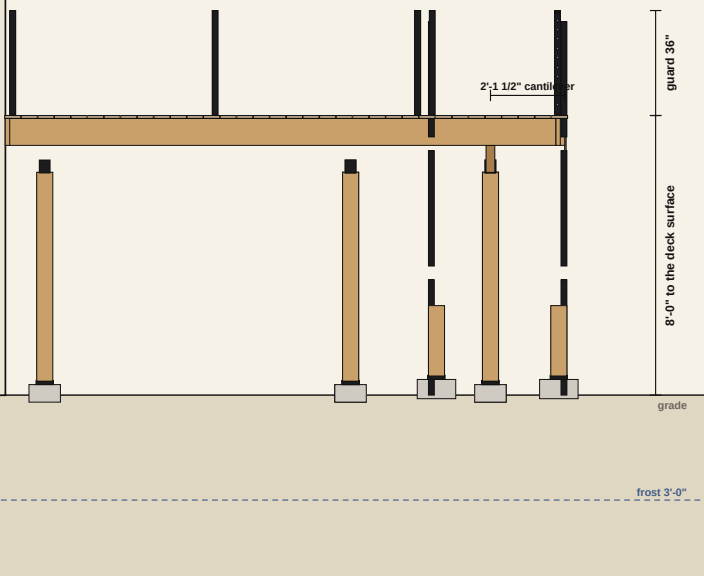
A-301 SECTION + FRONT ELEVATION

Cut through the deepest zone; heights to grade, frost line, cantilever, guard

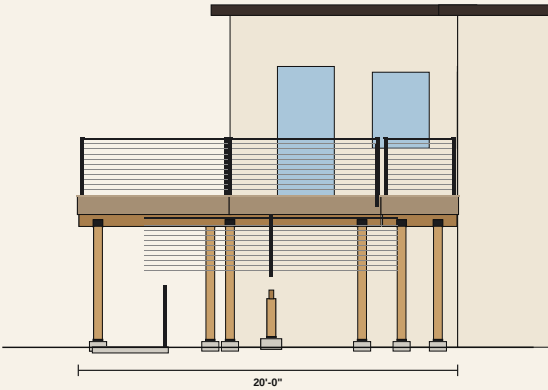
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SECTION through zone L at 2'-0" — looking toward the left end



FRONT ELEVATION — from the yard



SECTION NOTES

Deck surface 8'-0" above grade; guard 36" where the drop exceeds 30"
DROP (2)2x10 SYP N: 7'-11", 2 posts at 1'-0 1/2" / 6'-11 1/2" (worst 1,089 lb); DROP (2)2x10 SYP N+S: 15'-11", 3 posts at 1'-0 1/2" / 8'-0" / 14'-11 1/2" (worst 2,261 lb); DROP (2)2x10 Posts 9 x 6x6 on DP-50/50
2x10 ledger x 12'-0" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 13 fasteners
G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

Section left · elevation right · not to a common scale

Design intent per 2021 IRC R507 prescriptive. Field verify. Do not scale.

1 LEDGER AT THE HOUSE

membrane 12" up the wall
decking
flashing over the ledger, under the siding
2x10 joist in LUS28Z
house rim
ledger

2x10 ledger x 12'-0" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 13 fasteners

2 POST · BEAM · CAP

joists bear on the beam, H2.5AZ each
(2)2x10 drop beam
BC46Z
6x6 post

DROP (2)2x10 SYP N: 7'-11"; 2 posts at 1'-0 1/2" / 6'-11 1/2" (worst 1,089 lb); DROP (2)2x10 SYP N+S: 15'-11"; 3 posts at 1'-0 1/2" / 8'-0" / 14'-11 1

3 FOOTING · POST BASE

post
ABA66Z on the pier bolt
Diamond Pier head
grade
4 x 50" pins · DP-50/50

4 RAIL POST · GUARD

top rail
IRX post 36"
cables 3" OC · no bottom rail
rim (2-ply, inner ply pocketed)

IRX 36" — 9 posts (3 CORNER, 4 END, 2 LINE), 7 bays at 3'-7" / 5'-9 1/2" / 7'-7" / 7'-9 1/2" CTC (8' kits cut to bay), posts on the divider lines, no bottom rail; HeadLOK 6" x 4 per post

5 DECK EDGE · BORDER

border
field · 3/16" gaps
rim
fascia
overhang 1"

Cortex face screws with color-matched plugs: 2 per board at every joist and at both ends of every run, color-matched plugs (1442 field + 0 border/divider)

6 BLOCKING

Blocking one row over the beam, staggered up and down so every block can be end-nailed.

Schematic details, not to scale

Design intent per 2021 IRC R507 prescriptive. Field verify. Do not scale.



Plan from above

MATERIALS every piece, as ordered

Lumber

QTY	UNIT	ITEM	SKU
27	ea	2x10x12 #1 True Frame Joist - GC	210N12TFJ
7	ea	2x12x16 #1 True Frame Joist - GC	212N16TFJ
10	ea	2x10x16 #1 True Frame Joist - GC	210N16TFJ
3	ea	2x6x12 #1 KDAT - GC	26N12K
1	ea	2x4x12 #1 KDAT - GC	24N12K
1	ea	2x4x16 #1 KDAT - GC	24N16K
9	ea	6x6x8 #2 - GC	66NN8
3	ea	2x10x16 #1 True Frame Joist - GC (line 17)	210N16TFJ
4	ea	2x10x20 #1 True Frame Joist - GC	210N20TFJ

Footings

QTY	UNIT	ITEM	SKU
3	ea	Simpson Titen HD 316SS 3/8x3 Screw Anchor	THD37300H6SS
9	ea	Diamond Pier Deck Footing DP50	DP1000
9	ea	Simpson Adjustable Post Base 6x6 - Zmax	ABU66Z
11	ea	Quikrete 80# Concrete Mix	QT110180

Hardware

QTY	UNIT	ITEM	SKU
40	ea	Simpson 2x10 Shear Face Mount Hanger Zmax	LUS210Z
6	ea	Simpson 2x10 Double Concealed Face Mount Hanger Zmax	HUC2102-2Z
58	ea	Simpson Hurricane Ties Z-Max H2.5AZ	H2.5AZ
3	ea	Simpson Adjustable Post Cap 6x6 - Zmax	AC6Z
4	ea	Simpson 6x Post Cap Adjustable Zmax	LPC6Z
4	ea	Simpson 6x Heavy Adjustable Post Cap - Zmax	ACH6Z
4	ea	Simpson Deck Tension Tie Zmax with 1.5" screws	DTT2Z
1	box	FastenMaster LedgerLok Flat Head 5" (50 ct)	FMLLF005-50
1	ea	Camo 3" PRO-FIT Joist Hanger nail HDG (5 lb box)	0096175
1	ea	Camo Pro-Fit 1-1/2"x.148 Hanger nail PT SM 1M Positive Placement 100/PL	0606491

Flashing & waterproofing

QTY	UNIT	ITEM	SKU
6	ea	Nichigo G-Tape 3035 2"x65' Deck Flash Tape - Black w/ NO Liner	GT265DFT
2	ea	Nichigo G-Tape 3035 4"x65' Deck Flash Tape - Black w/ NO Liner	GT465DFT
1	ea	Nichigo G-Tape 3040 9"x65' Deck Flash Tape - Black w/ Liner	GT965DFT

Decking

QTY	UNIT	ITEM	SKU
10	ea	Azek Vintage 1x6x16 - Weathered Teak - Square	AZV1616WT
27	ea	Azek Vintage 1x6x20 - Weathered Teak - Square	AZV1620WT

Fasteners

QTY	UNIT	ITEM	SKU
3	ea	Cortex for TimberTech Vintage — 100 SF (screws + Weathered Teak plugs, setting tool)	CORTEX-100-WT
1	ea	Cortex for TimberTech Vintage — 300 SF (screws + Weathered Teak plugs, setting tool)	CORTEX-300-WT
2	ea	Cortex for TimberTech fascia — 30 LF (screws + Weathered Teak plugs)	CORTEX-FAS-WT
1	ea	Camo 3" x #9 Premium Deck Screw ProTech 2-Tan (350 ct)	0356174

Fascia

QTY	UNIT	ITEM	SKU
5	ea	Azek Vintage (1/2x12x12') Fascia - Weathered Teak	AZVFASCIA12WT

Rail

QTY	UNIT	ITEM	SKU
3	ea	Azek Impression Express 3x3x38" Horizontal Cable Post Kit, Corner - Black	AZTIXHC90PK38B
4	ea	Azek Impression Express 3x3x38" Horizontal Cable Post Kit, Level End - Black	AZTIXHCEPK38B
3	ea	Azek Impression Express 3x3x38" Horizontal Cable Post Kit, Level Inline - Black	AZTIXHCIPK38B
4	ea	Azek Impression Express 6' x 36"/42" Level Horizontal Cable Railing Section Kit - Black	AZTIXHCLP6B
3	ea	Azek Impression Express 8' x 36"/42" Level Horizontal Cable Railing Section Kit - Black	AZTIXHCLP8B
10	ea	Azek Impression Express 316ss Cable (20' x 1/8") & Hardware	AZTIXHC20CABLE
20	ea	Azek Impression Express 316ss Cable (40' x 1/8") & Hardware	AZTIXHC40CABLE
1	ea	Azek Impression Express Horizontal Cable Accessory Kit	AZTIXHCACCKIT
10	ea	Azek Impression Rail Express Classic Top Rail 6' - Black	AZTIXCSTR6B

QTY	UNIT	ITEM	SKU
1	ea	Azek Impression Rail Express Classic Top Rail 8' - Black	AZTIXCSTR8B
1	ea	Azek Impression Aluminum ADA Rail Pipe Rail and 90° Return Kit - Black	AZTIXADAPR90DRKB
1	ea	Azek Impression Aluminum ADA Rail ADA Pipe Rail Extension Kit - Black	AZTIXADAPREKB
1	ea	Azek Impression Express Touch-Up Paint / Install Tool Kit - Black	AZTIXACCKITB

Stairs

QTY	UNIT	ITEM	SKU
6	ea	Azek Vintage (3/4x8x12') Riser - Weathered Teak	AZVRISER12WT
6	ea	Simpson Adjustable Stair Stringer Z-Max	LSCZ
2	ea	Azek Impression Express 3x3x43" Horizontal Cable Post Kit, Stair Bottom End - Black	AZTIXHCESPK43B
4	ea	Azek Impression Express 3x3x43" Horizontal Cable Post Kit, Stair Inline - Black	AZTIXHCISPK43B
6	ea	Azek Impression Express 6' x 36"/42" Stair Horizontal Cable Railing Section Kit - Black	AZTIXHCSP6B

FASTENING & CONNECTORS

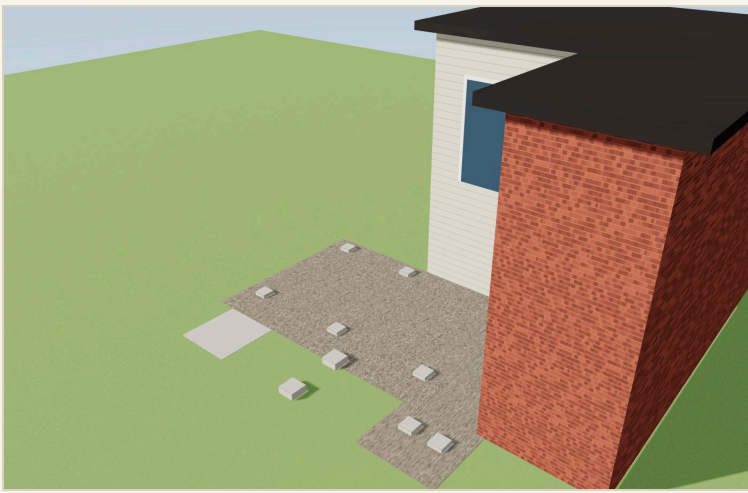
Ledger	2x10 ledger x 12'-0" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 13 fasteners
Joist tape	G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)
Field	Cortex face screws with color-matched plugs: 2 per board at every joist and at both ends of every run, color-matched plugs (1442 field + 0 border/divider)
Rail	IRX 36" — 9 posts (3 CORNER, 4 END, 2 LINE), 7 bays at 3'-7" / 5'-9 1/2" / 7'-7" / 7'-9 1/2" CTC (8' kits cut to bay), posts on the divider lines, no bottom rail; HeadLOK 6" x 4 per post

▼ Cut list

MEMBER	STOCK	LENGTH	QTY	NOTE
Field joists (N)	2x10	11'-6"	7	@ 12" OC, crown up
Side rim plies (N)	2x10	11'-6"	4	2-ply, laminated on the ground with RSS 2 rows @ 12" staggered
Front rim plies (N)	2x10	8'-0"	2	inner ply takes the hangers; outer ply laminated after
Rear rim plies (freestanding) (N)	2x10	8'-0"	2	
Blocking (N)	2x10	10 1/2"	18	3 rows at 1'-1 1/2", 9'-10 1/2", 5'-6" from the wall

MEMBER	STOCK	LENGTH	QTY	NOTE
Field boards run N	Vintage Weathered Teak	8'-1 15/16"	25	boards run parallel to the house
Rip at the house (N)	Vintage Weathered Teak	8'-1 15/16"	1	rip to 2.41"
Field joists (S)	2x10	11'-7 1/2"	7	@ 12" OC, crown up
Side rim plies (S)	2x10	11'-7 1/2"	4	2-ply, laminated on the ground with RSS 2 rows @ 12" staggered
Front rim plies (S)	2x10	8'-0"	2	inner ply takes the hangers; outer ply laminated after
Ledger (S)	2x10	8'-0"	1	top at deck height less board thickness; membrane behind, flashing over
Blocking (S)	2x10	10 1/2"	12	2 rows at 9'-10 1/2", 5'-0" from the wall
Field boards run S	Vintage Weathered Teak	8'-1 15/16"	25	boards run parallel to the house
Rip at the house (S)	Vintage Weathered Teak	8'-1 15/16"	1	rip to 2.41"
Field joists (L)	2x10	15'-7 1/2"	3	@ 12" OC, crown up
Side rim plies (L)	2x10	15'-7 1/2"	4	2-ply, laminated on the ground with RSS 2 rows @ 12" staggered
Front rim plies (L)	2x10	4'-0"	2	inner ply takes the hangers; outer ply laminated after
Ledger (L)	2x10	4'-0"	1	top at deck height less board thickness; membrane behind, flashing over
Blocking (L)	2x10	10 1/2"	4	2 rows at 13'-10 1/2", 7'-0" from the wall
Field boards run L	Vintage Weathered Teak	4'-1 15/16"	34	boards run parallel to the house
Stair stringers (step:S-L)	2x12	15'-1 11/16"	5	13 risers @ 7.385" · 12 treads @ 11.66" · stringers 12" OC
DROP (2)2x10 SYP N beam	2x10	7'-11"	2	y 1'-1 1/2"; posts at 1'-0 1/2" / 6'-11 1/2"
Posts under DROP (2)2x10 SYP N	6x6	6'-0"	2	field-measure each
DROP (2)2x10 SYP N+S beam	2x10	15'-11"	2	y 9'-10 1/2"; posts at 1'-0 1/2" / 8'-0" / 14'-11 1/2"
Posts under DROP (2)2x10 SYP N+S	6x6	6'-0"	3	field-measure each
DROP (2)2x10 SYP L beam	2x10	3'-11"	2	y 13'-10 1/2"; posts at 17'-0 1/2" / 18'-11 1/2"
Posts under DROP (2)2x10 SYP L	6x6	6'-0"	2	field-measure each
Stair carrier posts (step:S-L)	6x6	4'-0"	2	under the stringers at mid-run — field-measure

MEMBER	STOCK	LENGTH	QTY	NOTE
Stair carrier (step:S-L)	2x6	4'-6"	2	2 ply, notched stringers bear on it; hurricane tie each stringer
Stair treads (step:S-L)	Vintage Weathered Teak square	4'-0"	24	2 boards per tread, nosing 3/4" over the riser
Stair risers (step:S-L)	riser board	4'-0"	13	rip to 7.38" less tread thickness
Left fascia (n)	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Left fascia (n)	fascia	1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia (n)	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia (n)	fascia	1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Front fascia (n)	fascia	8'-1"	1	flush under the board nose; full board at the front corners, short piece at the house
Left fascia (s)	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Left fascia (s)	fascia	1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia (s)	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia (s)	fascia	1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Front fascia (s)	fascia	8'-1"	1	flush under the board nose; full board at the front corners, short piece at the house
Left fascia (l)	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Left fascia (l)	fascia	4'-0 1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia (l)	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia (l)	fascia	4'-0 1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Front fascia (l)	fascia	4'-1"	1	flush under the board nose; full board at the front corners, short piece at the house



Step 1 · Footings — 9 Diamond Pier DP-50/50 driven-pin footings

Step 1 · Footings

GOES IN

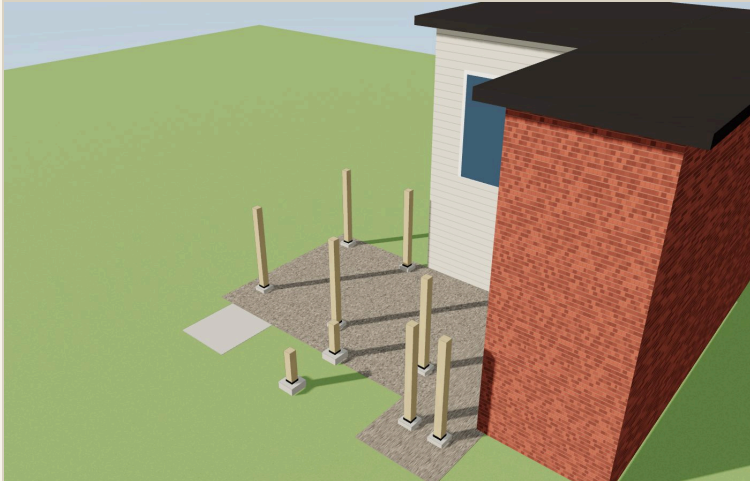
- (3) Simpson Titen HD 316SS 3/8x3 Screw Anchor
- (9) Diamond Pier Deck Footing DP50
- (9) Simpson Adjustable Post Base 6x6 - Zmax
- (11) Quikrete 80# Concrete Mix

HOW

- Call 811 and locate before any hole or pin. Strip the existing deck first.
- Pull the beam lines from the house wall:
DROP (2)2x10 SYP N at 1'-1 1/2" from the reference wall, posts at 1'-0 1/2" / 6'-11 1/2";
DROP (2)2x10 SYP N+S at 9'-10 1/2" from the reference wall, posts at 1'-0 1/2" / 8'-0" / 14'-11 1/2";
DROP (2)2x10 SYP L at 13'-10 1/2" from the reference wall, posts at 17'-0 1/2" / 18'-11 1/2".
- Set 9 Diamond Pier heads level at grade on undisturbed soil and drive the 4 pins in each; caps on.

CHECK

- Footings on the layout within 1/2"; bearing on undisturbed soil
 - Pins fully driven, heads level
 - Photos before backfill
-



Step 2 · Posts — 6x6 #2 GC on ZMAX bases

Step 2 · Posts

GOES IN

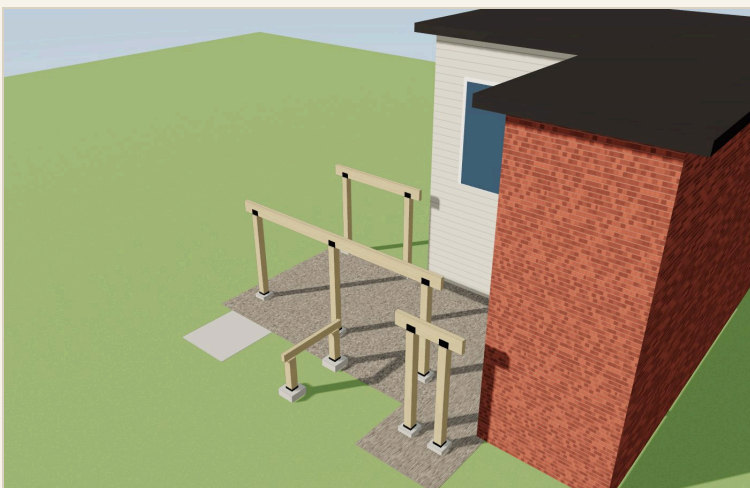
- (9) 6x6x8 #2 - GC
- (3) Simpson Titen HD 316SS 3/8x3 Screw Anchor
- (9) Simpson Adjustable Post Base 6x6 - Zmax

HOW

- 6x6 posts on the bases; cut so the beam top lands at the joist bottom (drop beam) or the joist top (flush beam): DROP (2)2x10 SYP N posts 6'-0" / DROP (2)2x10 SYP N+S posts 6'-0" / DROP (2)2x10 SYP L posts 6'-0" — field-measure each.
- Plumb both ways, brace, then the caps go on with the beams.

CHECK

- Posts plumb to a string; tops on the beam line
- No post in ground contact without a standoff



Step 4 · Beams — DROP (2)2x10 SYP N (2 posts) · DROP (2)2x10 SYP N+S (3 posts) · DROP (2)2x10 SYP L (2 posts)

Step 4 · Beams

GOES IN

- (3) 2x6x12 #1 KDAT - GC
- (1) 2x4x12 #1 KDAT - GC
- (9) 6x6x8 #2 - GC
- (3) Simpson Adjustable Post Cap 6x6 - Zmax
- (4) Simpson 6x Post Cap Adjustable Zmax
- (4) Simpson 6x Heavy Adjustable Post Cap - Zmax

HOW

- DROP (2)2x10 SYP N: 7'-11" one piece, 2 posts @ 5'-11" max
- DROP (2)2x10 SYP N+S: 15'-11" one piece, 3 posts @ 6'-11 1/2" max
- DROP (2)2x10 SYP L: 3'-11" one piece, 2 posts @ 1'-11" max
- Crown up, strung straight, caps screwed off with the SDS screws in the box.

CHECK

- Beam top = joist bottom (drop) / joist top (flush)
- Splices land on a post; caps fully fastened



Step 5 · Frame — 2x10 joists at 12", rims, ledger, blocking over the beam

Step 5 · Frame

GOES IN

- (27) 2x10x12 #1 True Frame Joist - GC
- (10) 2x10x16 #1 True Frame Joist - GC
- (3) 2x10x16 #1 True Frame Joist - GC (line 17)
- (4) 2x10x20 #1 True Frame Joist - GC
- (6) Nichigo G-Tape 3035 2"x65' Deck Flash Tape - Black w/ NO Liner
- (2) Nichigo G-Tape 3035 4"x65' Deck Flash Tape - Black w/ NO Liner
- (58) Simpson Hurricane Ties Z-Max H2.5AZ
- (1) Nichigo G-Tape 3040 9"x65' Deck Flash Tape - Black w/ Liner

HOW

- 2x10 ledger x 12'-0" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 13 fasteners
- N: 7 joists @ 12" OC x 11'-6"; S: 7 joists @ 12" OC x 11'-7 1/2"; L: 3 joists @ 12" OC x 15'-7 1/2". Crown up, hangers at the ledger.
- Blocking one row over the beam, staggered.
- G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

CHECK

- Frame square: diagonals within 1/8"
- Every hanger fully nailed; ledger fasteners per the schedule
- Tape continuous; photos of the ledger, hangers and footings before decking

Step 6 · Decking

GOES IN

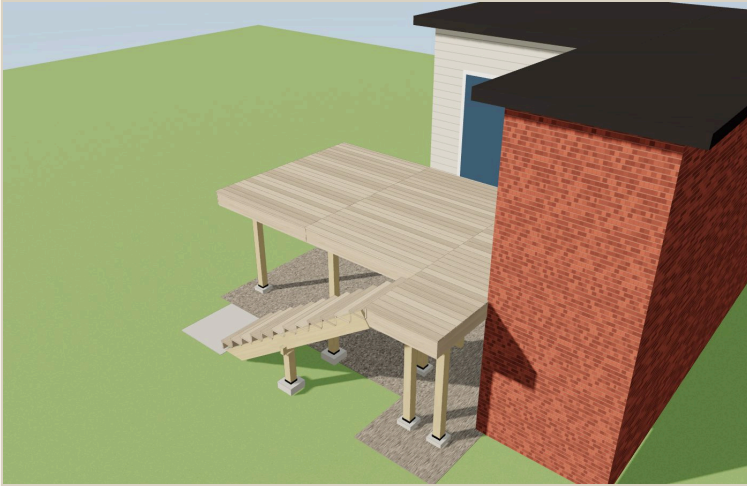
- (10) Azek Vintage 1x6x16 - Weathered Teak - Square
- (27) Azek Vintage 1x6x20 - Weathered Teak - Square
- (5) Azek Vintage (1/2x12x12') Fascia - Weathered Teak
- (3) Cortex for TimberTech Vintage — 100 SF (screws + Weathered Teak plugs, setting tool)
- (1) Cortex for TimberTech Vintage — 300 SF (screws + Weathered Teak plugs, setting tool)
- (2) Cortex for TimberTech fascia — 30 LF (screws + Weathered Teak plugs)
- (1) Camo 3" x #9 Premium Deck Screw ProTech 2-Tan (350 ct)

HOW

- Cortex face screws with color-matched plugs: 2 per board at every joist and at both ends of every run, color-matched plugs (1442 field + 0 border/divider)
- Borders first, then the field.
- Rows per zone: N 25 · S 25 · L 34. Borders and dividers first, then the field rows from the outer edge toward the house.

CHECK

- 3/16" gaps held
- No cut end shows at the outer edge
- Color: every board from the same run



Step 6 · Decking — TimberTech Vintage Weathered Teak, hidden fasteners

Step 7 · Rail

GOES IN

- (3) Azek Impression Express 3x3x38" Horizontal Cable Post Kit, Corner - Black
- (4) Azek Impression Express 3x3x38" Horizontal Cable Post Kit, Level End - Black
- (3) Azek Impression Express 3x3x38" Horizontal Cable Post Kit, Level Inline - Black
- (4) Azek Impression Express 6' x 36"/42" Level Horizontal Cable Railing Section Kit - Black
- (3) Azek Impression Express 8' x 36"/42" Level Horizontal Cable Railing Section Kit - Black
- (10) Azek Impression Express 316ss Cable (20' x 1/8") & Hardware
- (20) Azek Impression Express 316ss Cable (40' x 1/8") & Hardware

HOW

- IRX 36" — 9 posts (3 CORNER, 4 END, 2 LINE), 7 bays at 3'-7" / 5'-9 1/2" / 7'-7" / 7'-9 1/2" CTC (8' kits cut to bay), posts on the divider lines, no bottom rail; HeadLOK 6" x 4 per post

CHECK

- Top of rail 36" above the decking; posts plumb
- Cables tensioned / balusters under 4" clear
- End of construction



Step 7 · Rail — IRX 36" Black

INVESTMENT

QUOTE

CENTER AVE — 12 X 20 DECK WITH LANDING AND STAIR

2x10 SYP frame at 12", (2)2x10 beam, 6x6 posts — on 9 Diamond Pier footings.
TimberTech Vintage Weathered Teak decking, IRX cable rail.

264 SF · 3 zones · **8'-0"** above grade · **30 psf** snow design

Prepared for Alan · 11215 W Center Ave, Lakewood

PRICE

Total investment \$63,975 — financed, no money down

- 12 months no payments, no down payment
- 18 months no payments, no down payment
- 6.99% for 10 years — about \$742/month

Check or ACH: \$59,790 (7% savings)

WHAT THE PRICE INCLUDES

ITEM	VALUE
Demolition & haul-off	\$970
Foundation & structure	\$16,466
Decking & fascia	\$15,760
Railing	\$18,075
Stairs	\$8,519
Total, check or ACH	\$59,790
Total, financed	\$63,975

Engineering. GSX provides the engineer and pulls the permit. Engineering and the permit are billed at cost, separate from this price.

PLAN

3 zones · 264 SF · 20'-0" along the house · 8'-0" above grade · 44.0 LF Black rail

- **N · North bay — open end, no house behind it — 8'-0" x 12'-0".**
- **S · South bay — ledgered to the Hardie wall — 8'-0" x 12'-0".**
- **L · Stair landing — 4' bump-out at the brick corner — 4'-0" x 16'-0".**

Zones are named left to right standing in the yard facing the house.

- Stair step:S-L: 13 risers @ 7.38", 12 treads, 5 stringers, 4'-0" wide

NEXT STEP

This is a fixed-price proposal on the selections above. Check or ACH, or financed with approved credit. Numbers hold 30 days.



The interactive 3D model lives at the online version of this document.

GS Exterior Experts · Deck Division

Renders and drawings are design intent; dimensions on the sheets govern. Numbers hold 30 days.